

Sharad Patel

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Education

University of Florida, B.S. in Biomedical Engineering, Minor in Mathematics Expected May 2027

- GPA: 3.94/4.00
- Relevant Coursework: Differential Equations, Linear Algebra, Modeling in Mathematical Biology, Biomedical Instrumentation, Biosignals, Circuits, Biomaterials, Biomedical Transport Phenomena, Cellular Engineering Lab, Clinically Inspired Engineering Design, Biochemistry, Organic Chemistry, Genetics, C++ Programming

Research Experience

Undergraduate Research Assistant, Brunson Lab, University of Florida May 2026 – Present

- Develop a mechanistic model of primary graft dysfunction (PGD) after lung transplantation to identify its pathophysiological drivers and inform risk stratification.
- Conduct a systematic review of the PGD literature, synthesizing mechanistic and clinical findings to identify the gaps that guide model development.

Undergraduate Research Assistant, Holt Lab, University of Florida Dec 2025 – Present

- Develop a computer vision pipeline using the Segment Anything Model (SAM) to segment leaf images, generate per-leaf masks, and quantify percent infected tissue and structural damage through contour and convex-hull analysis.
- Convert pixel measurements to real-world area with reference-tag calibration and accelerate processing on Apple Silicon (PyTorch MPS), producing disease-severity metrics for large-scale ecological analysis.
- Build and validate ODE models of population and disease dynamics under environmental change, representing phenotypic plasticity with sigmoidal response functions.
- Extend the framework to include migration, coupling movement behavior with plasticity responses to capture spatial spread.

Undergraduate Research Assistant, Vedam-Mai Lab, University of Florida Sep 2025 – Present

- Lead author of a systematic review, *Immunotherapy in Parkinson's Disease*: design the search strategy, screen and extract studies, and synthesize evidence across active, passive, and cell-based immunotherapies.
- Investigate T-cell responses and neuroinflammation along the gut-brain axis, analyzing how immune markers and microbiome composition relate to disease progression.
- Contribute to experimental design discussions and prepare figures, research documentation, and lab presentations.

Undergraduate Research Assistant, Nichols Lab, University of Florida Aug 2024 – Present

- Collect fine-wire EMG and Vicon motion capture data during pinch and grasp tasks; process signals in Python to characterize movement patterns and muscle activation.
- Build subject-specific OpenSim thumb models personalized with ultrasound-derived muscle measurements; use inverse kinematics and static optimization to compare joint angles, moment arms, and muscle forces between healthy older adults and adults with thumb carpometacarpal osteoarthritis (CMC OA).
- Run sensitivity and uncertainty analyses on ultrasound-derived Hill-type parameters (optimal fiber length, tendon slack length, physiological cross-sectional area) to find the dominant drivers of personalization, and test whether a reduced Hill-type model reproduces the full simulation's effects.
- Analyze multi-muscle EMG recordings with paired *t*-tests and two-way ANOVA to compare activation across CMC OA stages.

Engineering Design

Senior Design Team Member, CONMED Distal Biceps Tendon Repair System Aug 2026 – Present

- Design a proof-of-concept distal biceps tendon repair system (implant, instruments, and sterile packaging) on an 8-person team for industry sponsor CONMED, to be evaluated by a sports medicine surgeon in cadaveric simulated use.
- Define 12 user needs in a design traceability matrix, with marginal and ideal target specifications grounded in biomechanical literature and FDA, ISO, and ASTM standards (e.g., ≥ 310 –440 N load to failure after 3,600 physiologic cycles, ≤ 3 mm tendon–bone gap, ≥ 5 mm nerve clearance).
- Benchmark cortical-button, suture-anchor, and bone-tunnel constructs and review IP and regulatory requirements, identifying a Class II 510(k) pathway (21 CFR 888.3040) and implant cost targets.

Surgical Research Team Captain, Dream Team Engineering, University of Florida Jan 2024 – Present

- Lead surgical simulation research: write IRB applications, study protocols, and statistical analysis plans, and design and run skills assessments for surgical residents.
- Work with transplant surgeons to develop realistic simulators for kidney and liver transplantation and infant IV placement.
- Share survey and experimental findings through team publications and conference presentations (see Publications & Presentations).

Circuits & Software Team Member, GRiP Gaming Team, University of Florida Aug 2023 – May 2024

- Co-developed the electrical and software systems for a 3D-printed prosthetic hand for a child with a below-elbow amputation.
- Programmed hand closure in response to biceps muscle (EMG) sensor signals using Arduino and C++.

Clinical Experience

Certified Nursing Assistant, HCA Florida North Florida Hospital – Gainesville, FL May 2026 – Present

- Provide direct patient care on a medical–surgical unit, including vital signs, activities of daily living, ambulation and repositioning, intake/output, and blood glucose monitoring.
- Document care in the electronic health record and promptly escalate changes in patient status to nursing staff.
- Support admissions, transfers, and discharges on a high-turnover unit, and assist with post-operative care, fall prevention, and infection control.

Teaching Experience

Teaching Assistant – Cellular Systems & Physiology, University of Florida Jan 2026 – Present

- Lead weekly human physiology labs, teaching students to apply engineering reasoning to interpret physiological data and build conceptual models.
- Redesign the lab curriculum (Summer–Fall 2026), restructuring activities and handouts around clear learning objectives and stronger conceptual scaffolding.
- Revise lab procedures and assessment questions to address recurring student errors, clarifying instructions and aligning lab work with lecture content.

Learning Assistant – Biomedical Transport Phenomena, University of Florida Aug 2026 – Present

- Hold weekly office hours and lead exam reviews on momentum, heat, and mass transport problem solving, targeting recurring conceptual difficulties.

Teaching Assistant – Organic Chemistry, University of Florida Jan 2026 – Aug 2026

- Held weekly office hours on reaction mechanisms, synthesis strategy, and spectroscopy; graded assessments and gave targeted feedback on recurring conceptual errors.

Math Tutor, Mathnasium – Tampa, FL Aug 2022 – Jun 2023

- Tutored a weekly cohort of 50 K–12 students, designing personalized learning plans for each student's needs.

Leadership & Service

Project Lead, BAPS Charities – Tampa, FL Aug 2022 – Present

- Lead community walkathons and health fairs: set goals and budgets, secure permits, coordinate 30–100+ volunteers, and manage sponsor outreach.
- Organize blood drives with local hospitals, managing logistics, screening workflows, and safety compliance to reduce wait times and increase donor throughput.

Regional Core Team Member, BAPS Swaminarayan Sanstha – Tampa, FL May 2021 – Present

- Design regional programs for 750–1,000 children (ages 6–13), and co-develop and deliver a weekly curriculum for 50–100 attendees with a team of six.

Co-Director of Internal Events, Dream Team Engineering, University of Florida May 2024 – May 2025

- Organized workshops, social events, and general body meetings, and ran large-scale events including a 24-Hour Design Challenge and a Shark Tank–style pitch competition.

Video Editor, Festival of Inspirations – Robbinsville, NJ Jun 2023 – Aug 2023

- Scripted and produced video content for a three-month cultural festival celebrating Hindu values and traditions.

Publications & Presentations

Evaluation of a Mixed-Reality Infant Caudal Epidural Block Simulator for Procedural Training Oct 2026

Sharad Patel, Pranav Kulkarni, Sanandan Ojha
Poster Presentation, 2026 BMES Annual Meeting – Orlando, FL

Evaluation of a 3D Kidney Transplant Simulator for Surgical Education: A Study Protocol Oct 2026

Sanandan Ojha, Thien-Nhi Nguyen, Shreya Sreekanth, **Sharad Patel**
Poster Presentation, 2026 BMES Annual Meeting – Orlando, FL

Modeling AI Model Collapse Using Markov Chains Apr 2026

Sharad Patel, Pranav Kulkarni, Sanandan Ojha
Oral Presentation, Undergraduate Mathematics Research Symposium, University of Florida – Gainesville, FL

Enhancing Anesthesia Monitoring with Computer Vision-Powered Train-of-Four Testing Apr 2025

Ansh Parikh, **Sharad Patel**, Vineet Khanolkar, Aditya Jithesh, Nishant Nagururu, Utsav Sharma
Poster Presentation, Engineering Research Symposium, PHHE

Effectiveness of 3D Box Model Trainers in Kidney Transplantation Training: A Controlled Study Mar 2025

John Cabrera, **Sharad Patel**, Makenna Myrick, Claudia Klejc, Ania Kelegama, Anne Titus
Presentation, 2024 Surgeons and Engineers: A Dialogue on Surgical Simulation

Improving Pediatric MRIs: A Prospective Study Using a Mock Scanner as a Replacement for Sedation Aug 2024

Shreya Mathur, Utsav Sharma, Ansh Parikh, Daisy Ann Titus, **Sharad Patel**
International Undergraduate Journal of Medicine, Disease, and Society

Projects

Mechanobiological Model of In-Stent Restenosis in Peripheral Stents Aug 2026 – Present

- Adapt a published reaction–diffusion model of restenosis (Escuer et al., 2019) from balloon-expandable coronary stents to self-expanding nitinol stents in the femoral artery. The model couples growth factors, MMP-2, extracellular matrix (ECM), and contractile and synthetic smooth muscle cells.
- Lead the problem definition and physiological rationale, linking the stent’s chronic outward force to smooth muscle cell migration and proliferation, ECM deposition, and neointimal hyperplasia.
- Added a chronic outward force module that links stent oversizing to arterial hoop stress. The re-implemented

base model matched published outputs (24.8% vs. $\sim 25\%$ restenosis at 300 days), and in a blind test against porcine data it predicted 73% area stenosis (observed: $64 \pm 17\%$).

- Tools: PDE modeling, numerical simulation, parameter calibration, sensitivity analysis

Height-Adjustable Axillary Crutch – CAD Design & Technical Report

Oct 2025

- Designed an 11-part telescoping pediatric crutch in Onshape, sized with the clinical rule that crutch length equals 77% of height, to fit users 1.33–1.56 m tall with ~ 180 mm of pin-locked height adjustment.
- Built a full assembly with fastened and slider mates and travel limits to simulate the telescoping leg and removable locking pins.
- Produced dimensioned multiview engineering drawings for every part, specifying ± 0.01 mm tolerances at sliding and pin interfaces and selecting materials for each part (aluminum alloy frame, ABS connector, TPU/rubber grips, stainless steel pins).
- Wrote a technical report covering design justification, part-by-part drawings, tolerancing rationale, and design limitations, with next steps including FEA load simulation and spring-loaded locking.
- Tools: Onshape, Fusion 360, tolerancing and technical drawing, technical writing

SCUDEM International Modeling Competition – *Outstanding* Award

- Modeled AI model collapse with two complementary frameworks: a Markov-chain ecosystem model for early-stage information loss and a layered discrete-distribution model for late-stage collapse.
- Simulated recursive human–AI training with Monte Carlo experiments and measured distribution drift using KL divergence and tail coverage, showing that model diversity stabilizes performance.
- Earned the *Outstanding* designation, the competition's highest award level, among international teams.
- Tools: MATLAB, probability theory, stochastic processes, information theory

COMAP Mathematical Contest in Modeling – Smartphone Battery Dynamics

- Developed a physics-based continuous-time ODE model that predicts smartphone battery state of charge under realistic use, accounting for processor load, display brightness, network activity, and thermal effects.
- Represented user activity with stochastic indicator functions and Fourier-series CPU and network profiles; a global Monte Carlo sensitivity analysis (10^4 simulations) identified processor load as the dominant driver of battery drain.
- Tools: MATLAB (ode45), stochastic modeling, uncertainty quantification

Spotted Seatrout Population Model & Management Simulator

GitHub

- Built a stock-assessment ODE model for Florida spotted seatrout combining logistic growth with fishing mortality, integrating FWC landings, catch-per-unit-effort, and spawning stock biomass (SSB) data from 2000–2023.
- Estimated parameters with constrained Nelder–Mead optimization and validated trajectories against independent SSB data ($< 10\%$ relative error); simulated 10-year constant- F harvest policies to identify F_{MSY} and B_{MSY} .
- Tools: Python (NumPy, pandas, SciPy, Matplotlib), Jupyter, Git

Neural Network from Scratch

- Implemented a feedforward neural network from first principles in NumPy, including forward propagation, backpropagation, and mini-batch gradient descent, without deep learning frameworks.
- Trained and evaluated the network on the MNIST handwritten digit dataset, tracking training and validation loss and accuracy.
- Tools: Python (NumPy, Matplotlib)

Bias Analysis of Student Organization Applications

- Analyzed multi-year application data to identify trends, predictors of acceptance, and sources of reviewer bias, including the effect of review order on final outcomes.
- Built scoring dashboards and recommended process changes (blinded first-pass review, balanced reviewer assignment, calibration sessions) to reduce bias and improve fairness.
- Tools: Python (NumPy, pandas, Matplotlib, Seaborn)

Connections Word Game – Full-Stack Web App

GitHub

- Built a full-stack clone of the NYT *Connections* puzzle, reproducing animated tile interactions, color-coded category reveals, and guess tracking.
- Implemented a persistent global leaderboard that tracks scores across sessions for competitive play.
- Tools: JavaScript, HTML, CSS

Personal Portfolio Website

sharad Patel.com

- Designed, built, and deployed a responsive portfolio site showcasing research, projects, and publications.
- Tools: React, JavaScript, HTML, CSS

Skills & Certifications

Programming: Python (NumPy, pandas, SciPy, PyTorch, OpenCV), MATLAB, C++, R, JavaScript, LaTeX

Modeling & Analysis: ODE/PDE modeling, Monte Carlo simulation, sensitivity and uncertainty analysis, parameter estimation, statistical testing (*t*-tests, ANOVA), computer vision

Engineering & Lab: SolidWorks, Onshape, Fusion 360, 3D printing, Arduino, OpenSim, Vicon motion capture, EMG, BSL-2 lab techniques

Research & Writing: Systematic reviews, IRB applications, study protocols, technical reports, scientific writing

Certifications: Certified Nursing Assistant (CNA), Basic Life Support (BLS)